

AN EFFICIENT ILQR ALGORITHM FOR OPTIMAL CONTROL WITH BOUNDED STATES AND INPUTS: APPLICATION TO SPACECRAFT MANEUVERS

Abhijeet*, Tarun Hejmadi[†], and Suman Chakravorty[‡]

This paper presents a constrained iterative Linear Quadratic Regulator (Box-iLQR) method for nonlinear aerospace optimal-control problems with bounded states and controls. Logarithmic barrier functions are used to enforce inequality constraints, while an outer continuation loop progressively reduces the barrier parameters. The resulting method preserves feasibility, improves the conditioning of the iLQR backward pass, and produces a time-varying feedback policy together with the nominal trajectory. The method is demonstrated on a spacecraft attitude maneuver and a minimum-fuel planar orbit transfer. For the attitude-control problem, Box-iLQR respects the actuator limits and stabilizes the maneuver under disturbance torques and measurement noise. For the orbit-transfer problem, a smooth minimum-fuel cost produces an approximate bang-off-bang thrust profile while satisfying the terminal conditions to high accuracy. Under control uncertainty, the feedback policy reduces the terminal radial-position error from order 10^{-4} to 10^{-7} and the radial-velocity error to approximately 10^{-6} . These results show that Box-iLQR provides constrained trajectory optimization and closed-loop stabilization within a single computational framework.

INTRODUCTION

Constrained trajectory optimization is fundamental to a broad range of aerospace guidance, navigation, and control problems. Representative applications include spacecraft attitude reorientation, rendezvous and proximity operations, low-thrust orbit transfer, planetary landing, launch vehicle guidance, and atmospheric entry. In these applications, the control trajectory must optimize a mission-dependent performance index while simultaneously satisfying nonlinear dynamics, prescribed boundary conditions, and path constraints arising from actuator saturation, pointing restrictions, collision avoidance, structural limits, and operational requirements. Because the governing dynamics are generally nonlinear and the constraints may become active over extended portions of the trajectory, analytical solutions are available only for a limited number of special cases. Consequently, reliable and computationally efficient numerical optimal-control methods are essential for both mission design and onboard trajectory generation.^{1–4}

Numerical optimal-control methods are commonly divided into indirect and direct approaches. Indirect methods apply the calculus of variations or Pontryagin’s minimum principle to derive the necessary conditions for optimality. The resulting state–costate boundary-value problem can provide highly accurate solutions and direct information about the optimal switching structure. How-

*Graduate Student, Aerospace Engineering, Texas A& M University, College Station, TX.

[†]Graduate Student, Aerospace Engineering, Texas A& M University, College Station, TX.

[‡]Professor, Aerospace Engineering, Texas A& M University, College Station, TX.

ever, the method may be sensitive to the initialization of the unknown costates, particularly for long-duration trajectories, strongly nonlinear dynamics, and problems containing discontinuous or nearly discontinuous controls. Direct methods instead discretize the state and control trajectories and transcribe the optimal-control problem into a finite-dimensional nonlinear program. These methods are generally easier to initialize and can accommodate a broad class of path and boundary constraints, but the resulting nonlinear program may become large as the number of discretization points increases.¹⁻³ These considerations have motivated considerable interest in trajectory-optimization algorithms that retain the computational structure of dynamic programming while avoiding the solution of a large, general-purpose nonlinear program.

Spacecraft attitude maneuvering provides a representative example of a nonlinear constrained optimal-control problem. Large-angle reorientation maneuvers involve nonlinear coupling between the attitude kinematics and rigid-body rotational dynamics, while practical actuators impose strict bounds on the available torque and angular momentum. The classical work of Vadali and Junkins considered both optimal open-loop maneuvers and stabilizing feedback control for rigid spacecraft.⁵ Nonlinear optimal-feedback approximations for large-angle maneuvers were subsequently developed by Carrington and Junkins.⁶ Time-optimal reorientation and the associated control-switching structure were investigated by Byers and Vadali,⁷ while Crassidis, Vadali, and Markley developed an optimal variable-structure approach for tracking large-angle spacecraft maneuvers.⁸ More recently, discrete-mechanics-based optimal-control formulations have been proposed to explicitly account for momentum and control constraints while preserving the underlying attitude geometry.⁹ These studies demonstrate the importance of incorporating actuator bounds into the maneuver-design process rather than applying saturation to an unconstrained control solution after optimization.

Fuel-optimal low-thrust orbit transfer presents a related but distinct computational challenge. The long transfer duration, oscillatory orbital dynamics, terminal constraints, and limited thrust authority can produce a highly sensitive optimization problem. Moreover, minimum-fuel formulations typically generate bang-off-bang thrust histories, for which the number and location of switching times are not known a priori. Homotopy methods have been used to continuously transform a smoother optimal-control problem into the desired minimum-fuel problem,¹⁰ while pseudospectral discretizations have been employed to obtain high-accuracy low-thrust solutions.¹¹ Taheri, Kolmanovsky, and Atkins developed a smoothing procedure for indirect minimum-fuel trajectory optimization that regularizes the discontinuous thrust law and progressively recovers the bang-off-bang solution.¹² Although these methods can produce accurate solutions, the treatment of nonsmooth controls, active control bounds, and robust closed-loop execution remains an important consideration for autonomous applications.

Differential dynamic programming and the iterative Linear Quadratic Regulator (iLQR) provide an alternative class of numerical optimal-control methods based on successive local approximations of the nonlinear problem. The underlying second-order trajectory-improvement principle was established in the early work of Mayne.¹³ The iLQR formulation uses a local linear approximation of the dynamics and a quadratic approximation of the cost to compute an improved control policy through alternating backward and forward passes.¹⁴ Because each forward rollout satisfies the nonlinear dynamics, iLQR is naturally suited to shooting-based trajectory optimization. Furthermore, the backward pass produces both a feedforward control correction and a time-varying state-feedback gain. The resulting policy can therefore be used not only to improve the nominal trajectory but also to stabilize its execution. The ability of this approach to generate and stabilize complex nonlinear motions has been demonstrated in online trajectory-optimization applications.¹⁵

Standard iLQR is formulated for unconstrained state and control spaces. Consequently, directly applying the algorithm to aerospace systems can produce controls that exceed actuator capabilities or trajectories that violate mission-critical state restrictions. Control-limited differential dynamic programming addresses box constraints on the control by solving a local bound-constrained quadratic program during the backward pass.¹⁶ However, the simultaneous treatment of state and control constraints is more difficult because state feasibility depends on the complete nonlinear forward propagation. Simple quadratic penalties do not guarantee feasibility and may require large penalty coefficients that degrade the conditioning of the local quadratic model.

In this work, state and control inequality constraints are incorporated into iLQR through logarithmic barrier functions. Interior-point and barrier methods are well-established tools in numerical optimization for treating inequality constraints by replacing the original constrained problem with a sequence of smooth unconstrained subproblems.¹⁷ For the present formulation, the barrier terms are included directly in the running and terminal costs. In addition to discouraging the trajectory from approaching an inadmissible region, the second derivatives of the logarithmic barriers contribute positive curvature to the local quadratic approximation used in the iLQR backward pass. This curvature acts as a natural regularization mechanism and is particularly useful for minimum-fuel problems in which the running cost may have little or no curvature over coast intervals.¹⁸

The resulting method, referred to herein as Box-iLQR, uses a nested solution procedure. For fixed barrier parameters, an inner iLQR loop performs backward and forward passes with regularization and line search until the corresponding barrier subproblem converges. An outer continuation loop then reduces the barrier parameters so that the sequence of strictly feasible solutions approaches a solution of the original constrained problem. The solution of each barrier subproblem is used to warm-start the next one. When a barrier reduction is too aggressive and causes the inner solve to fail, the reduction factor associated with the corresponding constraint is adjusted to obtain a more conservative continuation step. This adaptive relaxation strategy avoids requiring a single barrier schedule to be selected a priori for constraints with different scales and sensitivities.

An important feature of the proposed approach is that the time-varying feedback gains generated during the final iLQR backward pass are retained for closed-loop trajectory execution. Thus, the optimization does not return only an open-loop control history. It also provides a locally stabilizing affine policy around the optimized trajectory. This property is especially valuable for spacecraft applications, where environmental disturbances, actuator uncertainty, navigation errors, and imperfect modeling can cause an open-loop trajectory to depart substantially from its nominal solution.

The principal contributions of this work are summarized as follows:

1. A logarithmic-barrier iLQR formulation is presented for nonlinear optimal-control problems subject to componentwise state and control bounds.
2. An adaptive barrier-relaxation procedure is introduced to reduce sensitivity to the prescribed continuation schedule while maintaining a feasible warm start for successive barrier subproblems.
3. The time-varying feedback gains obtained from the converged trajectory optimization are used directly for closed-loop execution, without requiring a separate feedback-controller design.
4. The method is evaluated on two representative spacecraft problems: a fully actuated large-angle attitude maneuver with bounded control torques and a fuel-optimal orbit transfer with

bounded thrust magnitude and thrust direction. Closed-loop simulations are used to assess robustness to disturbance torques, measurement noise, and control uncertainty.

The remainder of this paper is organized as follows. First, the continuous- and discrete-time constrained optimal-control problems are defined. The logarithmic barrier formulation and the Box-iLQR backward and forward passes are then developed, followed by the adaptive barrier-relaxation algorithm. Numerical results are subsequently presented for the spacecraft attitude control and fuel-optimal orbit-transfer problems, including closed-loop robustness studies. Finally, the main conclusions and directions for future work are discussed.

BACKGROUND: THE OPTIMAL CONTROL PROBLEM

Consider the following optimal control problem:

$$\min_{\mathbf{u}(\cdot)} \quad \mathcal{J} = \phi(\mathbf{x}(t_f)) + \int_0^{t_f} c(\mathbf{x}(t), \mathbf{u}(t)) dt \quad (1a)$$

$$\text{s.t.} \quad \dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}(t), \mathbf{u}(t); t) \quad \forall t \in [0, t_f], \quad (1b)$$

$$\mathbf{x}(t) \in \mathcal{X} := \{\mathbf{x} \in \mathbb{R}^n \mid \underline{\mathbf{x}} \leq \mathbf{x} \leq \bar{\mathbf{x}}\}, \quad (1c)$$

$$\mathbf{u}(t) \in \mathcal{U} := \{\mathbf{u} \in \mathbb{R}^m \mid \underline{\mathbf{u}} \leq \mathbf{u} \leq \bar{\mathbf{u}}\}, \quad (1d)$$

$$\mathbf{x}(0) \text{ and } t_f \text{ are given.}$$

In the above equation, $\mathbf{x}(t) \in \mathbb{R}^n$ is the state and $\mathbf{u}(t) \in \mathbb{R}^m$ is the control input. The function $\mathbf{f} : \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R} \rightarrow \mathbb{R}^n$ describes the system dynamics. The objective functional $\mathcal{J}$ is composed of a terminal cost $\phi : \mathbb{R}^n \rightarrow \mathbb{R}$ and a running cost $c : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}$. The state and control are constrained to lie within the compact sets $\mathcal{X}$ and $\mathcal{U}$, respectively, for all $t \in [0, t_f]$.

The discrete counterpart of the problem mentioned in eq. (1) is:

$$\min_{\mathbf{u}_t} \quad \mathcal{J} = \Phi(\mathbf{x}_T) + \sum_{t=0}^{T-1} C(\mathbf{x}_t, \mathbf{u}_t), \quad (2a)$$

$$\text{s.t.} \quad \mathbf{x}_{t+1} = \mathbf{F}(\mathbf{x}_t, \mathbf{u}_t; t) \quad \forall t = 0, 1 \dots T-1, \quad (2b)$$

$$\mathbf{x}_t \in \mathcal{X} := \{\mathbf{x} \in \mathbb{R}^n \mid \underline{\mathbf{x}} \leq \mathbf{x} \leq \bar{\mathbf{x}}\}, \quad (2c)$$

$$\mathbf{u}_t \in \mathcal{U} := \{\mathbf{u} \in \mathbb{R}^m \mid \underline{\mathbf{u}} \leq \mathbf{u} \leq \bar{\mathbf{u}}\}, \quad (2d)$$

$$\mathbf{x}_0, T \text{ are given}$$

and $\Phi, C(\cdot)$, and $\mathbf{F}(\cdot, \cdot)$ are the discrete equivalents of ϕ, c and $\mathbf{f}$, respectively.

We can incorporate the inequality constraints into the cost function using a logarithmic barrier method. The constrained optimal control problem in (2) is thereby transformed into an unconstrained problem by augmenting the objective with barrier terms and the dynamic constraints with Lagrange multipliers. This leads to the formulation of the following barrier subproblem, where we

seek the stationary point of the augmented cost function $\mathcal{J}_{\mu,\sigma}$:

$$\begin{aligned} \mathcal{J}_{\mu,\sigma} = & \Phi(\mathbf{x}_T) - \sum_{i \in \mathcal{I}_x} \mu_i (\log(x_{T,i} - \underline{x}_i) + \log(\bar{x}_i - x_{T,i})) \\ & + \sum_{t=0}^{T-1} \left[C(\mathbf{x}_t, \mathbf{u}_t) - \sum_{i \in \mathcal{I}_x} \mu_i (\log(x_{t,i} - \underline{x}_i) + \log(\bar{x}_i - x_{t,i})) \right. \\ & \left. - \sum_{j \in \mathcal{I}_u} \sigma_j (\log(u_{t,j} - \underline{u}_j) + \log(\bar{u}_j - u_{t,j})) + \boldsymbol{\lambda}_{t+1}^\top (\mathbf{F}(\mathbf{x}_t, \mathbf{u}_t) - \mathbf{x}_{t+1}) \right]. \end{aligned} \quad (3)$$

In the expression above, the variables are defined as follows: $\boldsymbol{\mu} = [\mu_1, \mu_2, \dots, \mu_{n_1}]$ s.t. $\mu_i > 0 \ \forall i \in 1, \dots, n_1$ and $\boldsymbol{\sigma} = [\sigma_1, \sigma_2, \dots, \sigma_{m_1}]$ s.t. $\sigma_j > 0 \ \forall j \in 1, \dots, m_1$ are the barrier parameters corresponding to the state and control inequality constraints, respectively.

THEORY: THE BOX-ILQR ALGORITHM

The algorithm is initialized with a feasible trajectory. We note that finding such a trajectory can be a challenging problem in itself, but methods for doing so are outside the scope of this work. The core of the algorithm consists of the backward and forward passes, which we detail below.

Barrier Cost Terms and Their Derivatives Our cost function is augmented with logarithmic barrier terms to enforce inequality constraints. The running cost includes a barrier term $\omega(\mathbf{x}_t, \mathbf{u}_t)$, and the terminal cost includes a barrier term $\Omega(\mathbf{x}_T)$.

The **running barrier term** is defined as:

$$\begin{aligned} \omega(\mathbf{x}_t, \mathbf{u}_t) = & - \sum_{i \in \mathcal{I}_x} \mu_i (\log(x_{t,i} - \underline{x}_i) + \log(\bar{x}_i - x_{t,i})) \\ & - \sum_{j \in \mathcal{I}_u} \sigma_j (\log(u_{t,j} - \underline{u}_j) + \log(\bar{u}_j - u_{t,j})) \end{aligned} \quad (4)$$

Its first-order derivative (gradient) will have some zero and some non-zero elements. The second-order derivatives (Hessians) is a diagonal matrix, with some zero and some non-zero elements. The non-zero elements of gradient and Hessian are given by:

$$\begin{aligned} (\omega_{\mathbf{x}})_i &= -\mu_i \left(\frac{1}{x_{t,i} - \underline{x}_i} - \frac{1}{\bar{x}_i - x_{t,i}} \right), (\omega_{\mathbf{u}})_j = -\sigma_j \left(\frac{1}{u_{t,j} - \underline{u}_j} - \frac{1}{\bar{u}_j - u_{t,j}} \right) \\ (\omega_{\mathbf{xx}})_{ii} &= \mu_i \left(\frac{1}{(x_{t,i} - \underline{x}_i)^2} + \frac{1}{(\bar{x}_i - x_{t,i})^2} \right), \\ (\omega_{\mathbf{uu}})_{jj} &= \sigma_j \left(\frac{1}{(u_{t,j} - \underline{u}_j)^2} + \frac{1}{(\bar{u}_j - u_{t,j})^2} \right), \end{aligned} \quad (5)$$

where $i \in \mathcal{I}_x$ and $j \in \mathcal{I}_u$. The cross-term Hessian $\omega_{\mathbf{ux}}$ is zero.

The **terminal barrier term** is defined as:

$\Omega(\mathbf{x}_T) = - \sum_{i \in \mathcal{I}_x} \mu_i (\log(x_{T,i} - \underline{x}_i) + \log(\bar{x}_i - x_{T,i}))$. The non-zero elements of its derivatives with respect to the terminal state $\mathbf{x}_T$ are:

$$(\Omega_{\mathbf{x}})_i = -\mu_i \left(\frac{1}{x_{T,i} - \underline{x}_i} - \frac{1}{\bar{x}_i - x_{T,i}} \right) \text{ and } (\Omega_{\mathbf{xx}})_{ii} = \mu_i \left(\frac{1}{(x_{T,i} - \underline{x}_i)^2} + \frac{1}{(\bar{x}_i - x_{T,i})^2} \right).$$

Backward Pass The backward pass computes the local optimal control policy by propagating the coefficients of the updated costate $\lambda_t = \mathbf{S}_t \delta \mathbf{x}_t + \mathbf{v}_t$. It computes the factors, $\mathbf{v}_t$ and $\mathbf{S}_t$, backward in time from $t = T$ to $t = 0$.

1. Initialization ($t = T$): The value function derivatives are initialized at the terminal time using the derivatives of the total terminal cost, evaluated at the nominal state $\bar{\mathbf{x}}_T$: $\mathbf{v}_T = \Phi_{\mathbf{x}} + \Omega_{\mathbf{x}}$ and $\mathbf{S}_T = \Phi_{\mathbf{xx}} + \Omega_{\mathbf{xx}}$.

2. Backward Recursion ($t = T - 1, \dots, 0$): For each time step, we compute the derivatives of the action-value function Q_t for deviations around $(\bar{\mathbf{x}}_t, \bar{\mathbf{u}}_t)$:

$$Q_{\mathbf{x}} = (C_{\mathbf{x}} + \omega_{\mathbf{x}}) + \mathbf{F}_{\mathbf{x}}^{\top} \mathbf{v}_{t+1} \quad (6)$$

$$Q_{\mathbf{u}} = (C_{\mathbf{u}} + \omega_{\mathbf{u}}) + \mathbf{F}_{\mathbf{u}}^{\top} \mathbf{v}_{t+1} \quad (7)$$

$$Q_{\mathbf{xx}} = (C_{\mathbf{xx}} + \omega_{\mathbf{xx}}) + \mathbf{F}_{\mathbf{x}}^{\top} \mathbf{S}_{t+1} \mathbf{F}_{\mathbf{x}} \quad (8)$$

$$Q_{\mathbf{uu}} = (C_{\mathbf{uu}} + \omega_{\mathbf{uu}}) + \mathbf{F}_{\mathbf{u}}^{\top} \mathbf{S}_{t+1} \mathbf{F}_{\mathbf{u}} \quad (9)$$

$$Q_{\mathbf{ux}} = C_{\mathbf{ux}} + \mathbf{F}_{\mathbf{u}}^{\top} \mathbf{S}_{t+1} \mathbf{F}_{\mathbf{x}} \quad (10)$$

All derivatives are evaluated at the nominal trajectory, $(\bar{\mathbf{x}}_t, \bar{\mathbf{u}}_t)$. The feed-forward term $\mathbf{k}_t$ and feedback gain matrix $\mathbf{K}_t$ are then computed. A regularization term $\zeta \mathbf{I}$ is added to $Q_{\mathbf{uu}}$ to ensure it is positive-definite.

$$\mathbf{k}_t = -(Q_{\mathbf{uu}} + \zeta \mathbf{I})^{-1} Q_{\mathbf{u}} \quad (11)$$

$$\mathbf{K}_t = -(Q_{\mathbf{uu}} + \zeta \mathbf{I})^{-1} Q_{\mathbf{ux}} \quad (12)$$

Finally, the values of $\mathbf{S}_t$ and $\mathbf{v}_t$ are updated:

$$\mathbf{v}_t = Q_{\mathbf{x}} + \mathbf{K}_t^{\top} Q_{\mathbf{uu}} \mathbf{k}_t + \mathbf{K}_t^{\top} Q_{\mathbf{u}} + Q_{\mathbf{ux}}^{\top} \mathbf{k}_t \quad (13)$$

$$\mathbf{S}_t = Q_{\mathbf{xx}} + \mathbf{K}_t^{\top} Q_{\mathbf{uu}} \mathbf{K}_t + \mathbf{K}_t^{\top} Q_{\mathbf{ux}} + Q_{\mathbf{ux}}^{\top} \mathbf{K}_t \quad (14)$$

Forward Pass In the forward pass, a new trajectory is generated by applying the computed control policy. A line search is performed with a parameter $\alpha \in (0, 1]$ to ensure improvement in the cost function and maintain feasibility. Starting from $\mathbf{x}_0^{\text{new}} = \bar{\mathbf{x}}_0$:

$$\mathbf{u}_t^{\text{new}} = \bar{\mathbf{u}}_t + \alpha \mathbf{k}_t + \mathbf{K}_t (\mathbf{x}_t^{\text{new}} - \bar{\mathbf{x}}_t) \quad t \in [0, T - 1] \quad (15)$$

$$\mathbf{x}_{t+1}^{\text{new}} = \mathbf{F}(\mathbf{x}_t^{\text{new}}, \mathbf{u}_t^{\text{new}}) \quad t \in [0, T - 1] \quad (16)$$

The new trajectory $(\mathbf{x}^{\text{new}}, \mathbf{u}^{\text{new}})$ becomes the nominal trajectory for the next iteration. This forward-backward procedure is repeated until convergence. The backtracking line search on α ensures that the augmented cost function $\mathcal{J}_{\mu, \sigma}$ decreases at each iteration. A new trajectory is accepted if it provides a sufficient decrease, typically measured by comparing the actual reduction in cost to the reduction predicted by the local quadratic model. This ensures the algorithm makes progress towards a local minimum. A common acceptance criterion is when the ratio of actual to expected reduction is positive:

$$\frac{\mathcal{J}(\text{new}) - \mathcal{J}(\text{old})}{\Delta \mathcal{J}_{\text{expected}}} > c_1 \quad (17)$$

where $\mathcal{J}(\text{old})$ and $\mathcal{J}(\text{new})$ are the costs of the old and new trajectories, $c_1 \in (0, 1)$ is a small constant (e.g., 10^{-4}), and $\Delta \mathcal{J}_{\text{expected}} < 0$ is the cost change predicted by the quadratic approximation.

$$\Delta \mathcal{J}_{\text{expected}} = -\left(\alpha - \frac{\alpha^2}{2}\right) \sum_{i=0}^{T-1} Q_{\mathbf{u}}^{\top} Q_{\mathbf{uu}}^{-1} Q_{\mathbf{u}}. \quad (18)$$

One should note here that we need Q_{uu} to be positive definite for $\Delta J_{\text{expected}} < 0$, i.e, the expected cost to decrease.

Algorithm 1 Box-iLQR

Require: Initial feasible trajectory $(\tilde{\mathbf{x}}, \tilde{\mathbf{u}})$
Require: Initial barrier parameters $\boldsymbol{\mu}_0, \boldsymbol{\sigma}_0$
Require: Initial reduction factors $\mathbf{r}_\mu \in (0, 1)^{n_1}, \mathbf{r}_\sigma \in (0, 1)^{m_1}$
Require: Reduction factor update rate $\beta_r > 1$ (e.g., 1.5)
Require: Barrier termination tolerance $\epsilon_{\text{barrier}}$

```

1: Initialize  $\boldsymbol{\mu} \leftarrow \boldsymbol{\mu}_0, \boldsymbol{\sigma} \leftarrow \boldsymbol{\sigma}_0$ 
2: Initialize  $\boldsymbol{\mu}_{\text{prev}} \leftarrow \boldsymbol{\mu}, \boldsymbol{\sigma}_{\text{prev}} \leftarrow \boldsymbol{\sigma}$ 
3: while  $\|\boldsymbol{\mu}\| > \epsilon_{\text{barrier}}$  or  $\|\boldsymbol{\sigma}\| > \epsilon_{\text{barrier}}$  do
4:                                      $\triangleright$  Solve the inner-loop problem with a warm start.
5:    $(\bar{\mathbf{x}}, \bar{\mathbf{u}}), \text{success}, \text{failed\_idx} \leftarrow \text{iLQR.SOLVE}(\bar{\mathbf{x}}, \bar{\mathbf{u}}, \boldsymbol{\mu}, \boldsymbol{\sigma})$ 
6:   if not success then
7:                                      $\triangleright$  Reduction was too aggressive; revert and adapt.
8:      $\boldsymbol{\mu} \leftarrow \boldsymbol{\mu}_{\text{prev}}, \boldsymbol{\sigma} \leftarrow \boldsymbol{\sigma}_{\text{prev}}$ 
9:     if constraint  $i$  in  $\boldsymbol{\mu}$  failed (from failed_idx) then
10:       $(\mathbf{r}_\mu)_i \leftarrow \min(1, (\mathbf{r}_\mu)_i \cdot \beta_r)$   $\triangleright$  Make reduction less aggressive.
11:     else if constraint  $j$  in  $\boldsymbol{\sigma}$  failed then
12:       $(\mathbf{r}_\sigma)_j \leftarrow \min(1, (\mathbf{r}_\sigma)_j \cdot \beta_r)$ 
13:     end if
14:     continue  $\triangleright$  Retry solving with old parameters and new rates.
15:   else
16:                                      $\triangleright$  Successful solve; prepare for next reduction.
17:      $\boldsymbol{\mu}_{\text{prev}} \leftarrow \boldsymbol{\mu}, \boldsymbol{\sigma}_{\text{prev}} \leftarrow \boldsymbol{\sigma}$ 
18:      $\boldsymbol{\mu} \leftarrow \mathbf{r}_\mu \odot \boldsymbol{\mu}$   $\triangleright$  Element-wise multiplication.
19:      $\boldsymbol{\sigma} \leftarrow \mathbf{r}_\sigma \odot \boldsymbol{\sigma}$ 
20:   end if
21: end while
22: return Optimal trajectory  $(\tilde{\mathbf{x}}, \tilde{\mathbf{u}})$ 

```

Barrier Relaxation The Box-iLQR method presented in the previous section solves the optimization problem for a *fixed* set of barrier parameters $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$. To recover the solution to the original constrained problem, these parameters must be driven to zero. This is managed by an outer loop that iteratively reduces the barrier parameters and uses the iLQR solver to find the solution for each new subproblem. This sequence of solutions ideally follows the *central path* towards the constrained optimum. The complete procedure, which we refer to as the Box-iLQR algorithm, is detailed in Algorithm 1. The process begins with conservatively large initial barrier parameters, $\boldsymbol{\mu}_0$ and $\boldsymbol{\sigma}_0$, and a feasible initial trajectory. The algorithm then enters an inner loop where it first solves the iLQR subproblem for the current parameters, using the solution from the previous iteration as a warm start to accelerate convergence.

A crucial aspect of this process is the parameter update rule. After a successful inner-loop solve, each component of $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$ is decreased by a multiplicative factor. A naive or overly aggressive reduction can push the warm-start trajectory into an infeasible region where the logarithmic barriers

are undefined, causing the inner-loop solver to fail. To handle this, our update schedule is adaptive. If the iLQR solver fails to converge, it indicates that the barrier parameters were reduced too severely. In response, the algorithm reverts the parameter update and applies a more conservative reduction factor for the corresponding constraint in the next attempt. This process continues until the barrier parameters are below a specified tolerance, ensuring their contribution to the total cost is negligible and the solution closely approximates the true constrained optimum.

Feedback Stabilization

In addition to generating a nominal optimal trajectory, the iLQR backward pass provides a time-varying feedback gain matrix at every time step. After convergence, these gains define a local linear time-varying tracking controller about the optimal trajectory,

$$\mathbf{u}_k = \mathbf{u}_k^* + \mathbf{K}_k^* (\mathbf{x}_k - \mathbf{x}_k^*), \quad (19)$$

where $\mathbf{x}_k^*$ and $\mathbf{u}_k^*$ denote the nominal state and control trajectories. The logarithmic barriers make the resulting feedback policy inherently constraint-aware: as a control component approaches its saturation boundary, the corresponding barrier curvature increases and the associated row of $\mathbf{K}_k^*$ diminishes, thereby reducing feedback corrections through the saturated channel. Similarly, information associated with an approaching state constraint is propagated backward through the value function recursion and influences the feedback gains at preceding time steps.¹⁸ Thus, the gains obtained during trajectory optimization can also be used to reject small disturbances and correct deviations from the nominal trajectory without requiring a separate feedback-controller design. In the Results section, we use these converged gains to perform closed-loop stabilization of the spacecraft attitude and orbit-transfer trajectories in the presence of process disturbances, measurement noise, and control uncertainty.

NUMERICAL RESULTS

The proposed Box-iLQR method is evaluated on two nonlinear spacecraft optimal-control problems: a fully actuated spacecraft attitude maneuver and a fuel-optimal orbit transfer. In both examples, Box-iLQR is first used to obtain a locally optimal nominal state–control trajectory satisfying the prescribed box constraints. The time-varying feedback gains obtained from the final backward pass are then retained and used for closed-loop propagation in the presence of process disturbances, control uncertainty, and measurement noise.

Let

$$\{\bar{\mathbf{x}}_k, \bar{\mathbf{u}}_k, \mathbf{K}_k\}_{k=0}^{N-1} \quad (20)$$

denote the nominal trajectory, nominal control, and feedback gains obtained at convergence. The closed-loop control law used in the robustness studies is

$$\mathbf{u}_k^{\text{cl}} = \Pi_{\mathcal{U}} [\bar{\mathbf{u}}_k + \mathbf{K}_k (\hat{\mathbf{x}}_k - \bar{\mathbf{x}}_k)], \quad (21)$$

where $\hat{\mathbf{x}}_k$ is the measured state and $\Pi_{\mathcal{U}}[\cdot]$ denotes projection onto the admissible control box. Consequently, the control limits remain satisfied during the closed-loop simulations even when the state is displaced from the nominal trajectory.

Spacecraft Attitude Control

Problem formulation The attitude state is defined as

$$\mathbf{x} = [\phi \quad \theta \quad \psi \quad \omega_1 \quad \omega_2 \quad \omega_3]^T, \quad (22)$$

where (ϕ, θ, ψ) are the 3–2–1 Euler angles and $\boldsymbol{\omega} = [\omega_1, \omega_2, \omega_3]^T$ is the angular velocity expressed in the spacecraft body frame. The attitude kinematics and rotational dynamics are

$$\begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} 1 & \sin \phi \tan \theta & \cos \phi \tan \theta \\ 0 & \cos \phi & -\sin \phi \\ 0 & \frac{\sin \phi}{\cos \theta} & \frac{\cos \phi}{\cos \theta} \end{bmatrix} \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix}, \quad (23)$$

and

$$\dot{\boldsymbol{\omega}} = \mathbf{I}^{-1} (\mathbf{u} - \boldsymbol{\omega} \times \mathbf{I} \boldsymbol{\omega}), \quad (24)$$

respectively. The spacecraft is fully actuated, and its inertia matrix is

$$\mathbf{I} = \text{diag} (140, 120, 130). \quad (25)$$

The three control torques are subject to the componentwise bounds

$$-10^{-3} \leq u_{i,k} \leq 10^{-3}, \quad i = 1, 2, 3. \quad (26)$$

The desired terminal attitude and angular velocity are

$$\boldsymbol{\eta}_f = [\pi \quad -\pi/4 \quad -\pi/2]^T, \quad \boldsymbol{\omega}_f = \mathbf{0}, \quad (27)$$

where $\boldsymbol{\eta} = [\phi, \theta, \psi]^T$. The maneuver is discretized using a zero-order hold on the control and a fourth-order Runge–Kutta integration scheme. A time step of

$$\Delta t = 1 \text{ s} \quad (28)$$

and a final time of

$$t_f = 3000 \text{ s} \quad (29)$$

are used, resulting in $N = 3000$ control intervals.

The attitude maneuver is obtained by minimizing a quadratic objective of the form

$$J = \frac{1}{2} \mathbf{e}_N^T \mathbf{Q}_f \mathbf{e}_N + \sum_{k=0}^{N-1} \left[\frac{1}{2} \mathbf{e}_k^T \mathbf{Q} \mathbf{e}_k + \frac{1}{2} \mathbf{u}_k^T \mathbf{R} \mathbf{u}_k \right] \Delta t, \quad (30)$$

where $\mathbf{e}_k$ is the state error relative to the desired terminal attitude and zero angular velocity. $\mathbf{Q}, \mathbf{R} = 5I$ and $\mathbf{Q}_f = 50000I$. The Euler-angle components of the error are computed modulo 2π so that they correspond to the minimum angular difference.

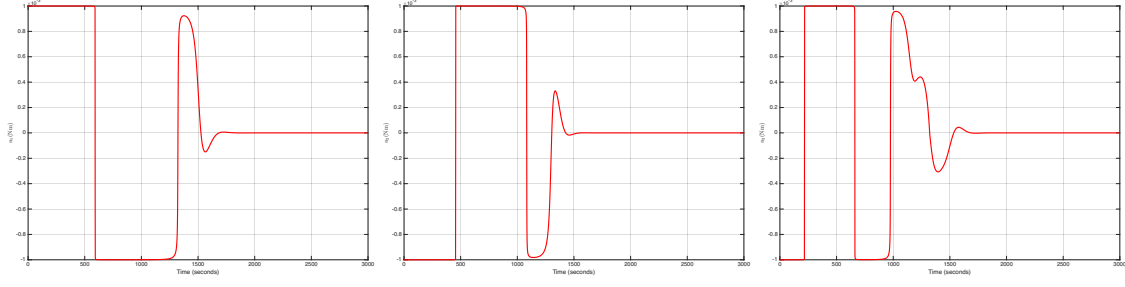


Figure 1: Optimal attitude-control torque histories obtained using Box-iLQR. From left to right, the panels show u_1 , u_2 , and u_3 . The torque histories remain within the componentwise bounds $|u_i| \leq 10^{-3}$ and contain extended boundary arcs.

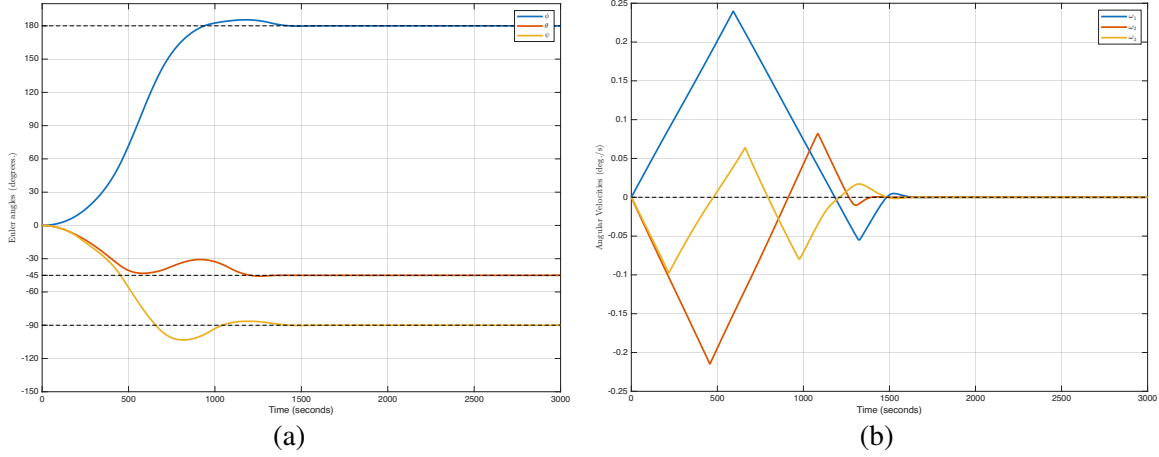


Figure 2: Nominal attitude trajectory: (a) Euler angles and (b) body-frame angular velocities.

Nominal constrained solution Figure 1 shows the three control-torque histories. The solution exhibits extended intervals of active control constraints, demonstrating that the box constraints directly influence the optimal maneuver. The first control torque remains at its upper bound for approximately the first 600 s and subsequently remains at its lower bound until approximately 1300 s. It then approaches zero and has effectively settled by approximately 1800 s. The second control initially remains at its lower bound for approximately 450 s, switches to its upper bound until approximately 1000 s, and approaches zero by approximately 1500 s. The third control displays a more complex switching structure. It initially remains at its lower bound for approximately 200 s, switches to its upper bound until approximately 700 s, returns briefly to its lower bound, and then approaches zero by approximately 1700 s.

The corresponding Euler-angle histories are presented in Fig. 2(a). The roll, pitch, and yaw angles converge to π , $-\pi/4$, and $-\pi/2$, respectively, before approximately 1500 s. The angular-velocity histories in Fig. 2(b) exhibit approximately linear growth and decay over several portions of the maneuver. These intervals correspond to the extended periods during which the control torques remain on their bounds. All three angular velocities subsequently converge to zero, with the dominant transient behavior disappearing by approximately 1600 s.

Closed-loop robustness under disturbances and measurement noise The robustness of the time-varying feedback policy is evaluated by perturbing both the spacecraft dynamics and the measured state. The disturbance torque is modeled independently along each body axis as

$$\boldsymbol{\tau}_d(t) = \boldsymbol{\xi} + \mathbf{a} \sin(\boldsymbol{\nu}t) + \mathbf{b} \cos(\boldsymbol{\nu}t), \quad (31)$$

where the trigonometric operations are applied componentwise. The stochastic variables are independently sampled according to

$$\xi_i \sim \mathcal{N}(0, (10^{-5})^2), \quad (32)$$

$$a_i, b_i \sim \mathcal{N}(0, (10^{-6})^2), \quad (33)$$

$$\nu_i \sim \mathcal{N}(0, (10^{-4})^2), \quad i = 1, 2, 3. \quad (34)$$

The perturbed angular-velocity dynamics therefore become

$$\dot{\boldsymbol{\omega}} = \mathbf{I}^{-1} [\mathbf{u} + \boldsymbol{\tau}_d(t) - \boldsymbol{\omega} \times \mathbf{I}\boldsymbol{\omega}]. \quad (35)$$

Additive measurement noise is introduced according to

$$\hat{\boldsymbol{\eta}}_k = \boldsymbol{\eta}_k + \mathbf{n}_{\eta,k}, \quad \mathbf{n}_{\eta,k} \sim \mathcal{N}(\mathbf{0}, (10^{-4})^2 \mathbf{I}_3), \quad (36)$$

and

$$\hat{\boldsymbol{\omega}}_k = \boldsymbol{\omega}_k + \mathbf{n}_{\omega,k}, \quad \mathbf{n}_{\omega,k} \sim \mathcal{N}(\mathbf{0}, (10^{-6})^2 \mathbf{I}_3). \quad (37)$$

These disturbance and measurement levels provide a representative test of the controller under simultaneous actuator-side disturbances and sensor uncertainty.

Figure 3 compares the nominal trajectory with a disturbed trajectory when the nominal control history is applied without feedback. The red curves represent the nominal states, whereas the black curves represent the disturbed states. The attitude-angle deviations are more pronounced than the angular-velocity deviations; nevertheless, observable departures from the nominal trajectory occur in all six states. The deviations accumulate because the open-loop control cannot compensate for disturbances that occur after the trajectory has been computed.

The same disturbance realizations are subsequently propagated using the time-varying feedback gains obtained from Box-iLQR. The results are shown in Fig. 4. The red curves denote the nominal solution, while the dotted black curves denote three representative perturbed closed-loop trajectories. The feedback policy drives each perturbed trajectory back toward the nominal solution and prevents the state errors from accumulating over the duration of the maneuver. In particular, the terminal attitude and angular-velocity errors remain small even though the controller is simultaneously subjected to process disturbances and noisy state measurements.

A Monte Carlo study with $N_{MC} = 100$ independent noise realizations is used to quantify the improvement obtained from feedback. Let

$$\mathbf{e}_{\eta,k}^{(j)} = \text{wrap}(\boldsymbol{\eta}_k^{(j)} - \bar{\boldsymbol{\eta}}_k) \quad (38)$$

and

$$\mathbf{e}_{\omega,k}^{(j)} = \boldsymbol{\omega}_k^{(j)} - \bar{\boldsymbol{\omega}}_k \quad (39)$$

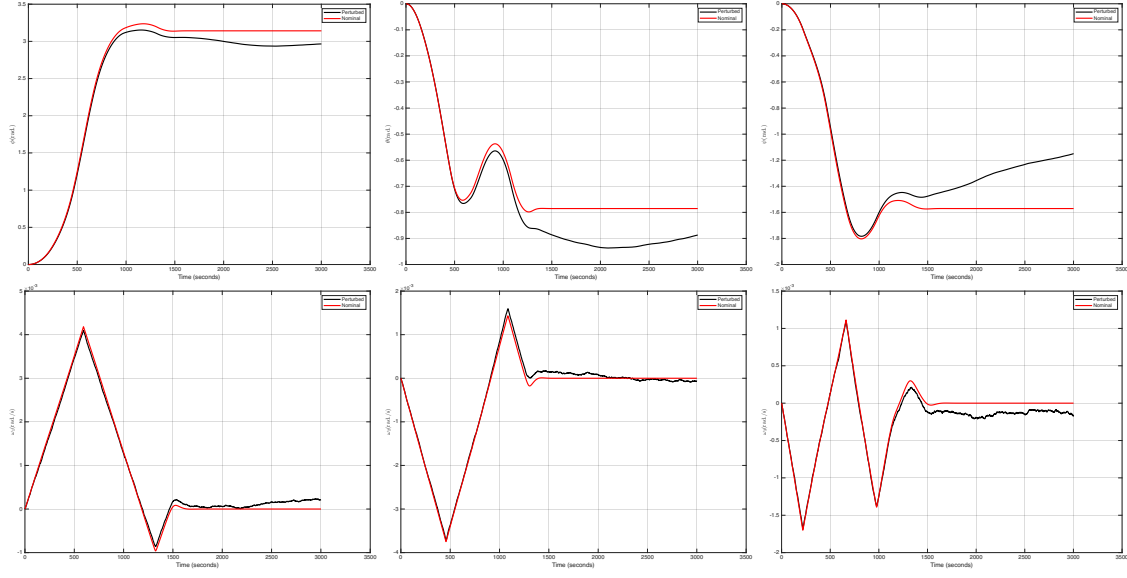


Figure 3: Attitude response under disturbances and measurement noise without feedback. The six panels correspond to the three Euler angles and three body-frame angular velocities. The nominal trajectory is shown in red and the disturbed open-loop trajectory is shown in black.

Table 1: Attitude-control Monte Carlo errors for 100 disturbance and measurement-noise realizations. The entries shown here are placeholders and must be replaced by the final simulation results.

Error quantity	Without feedback		With feedback	
	Mean	Maximum	Mean	Maximum
Euler-angle error [rad]	0.0620	0.0702	4.68×10^{-5}	1.06×10^{-4}
Angular-rate error [rad/s]	3.12×10^{-5}	3.34×10^{-5}	1.05×10^{-5}	2.21×10^{-5}

denote the attitude and angular-velocity errors for Monte Carlo realization j . The mean and maximum trajectory errors are defined as

$$e_{z,\text{mean}} = \frac{1}{N_{\text{MC}}} \sum_{j=1}^{N_{\text{MC}}} \left\| \mathbf{e}_z^{(j)} \right\|_2, \quad (40)$$

and

$$e_{z,\text{max}} = \max_{j=1,\dots,N_{\text{MC}}} \left\| \mathbf{e}_z^{(j)} \right\|_2, \quad z \in \{\eta, \omega\}. \quad (41)$$

The representative values in Table 1 illustrate the expected trend: the open-loop trajectories experience persistent deviations from the nominal maneuver, whereas the Box-iLQR feedback policy reduces both the mean and worst-case errors. The final manuscript will report the values obtained directly from the complete Monte Carlo data set.

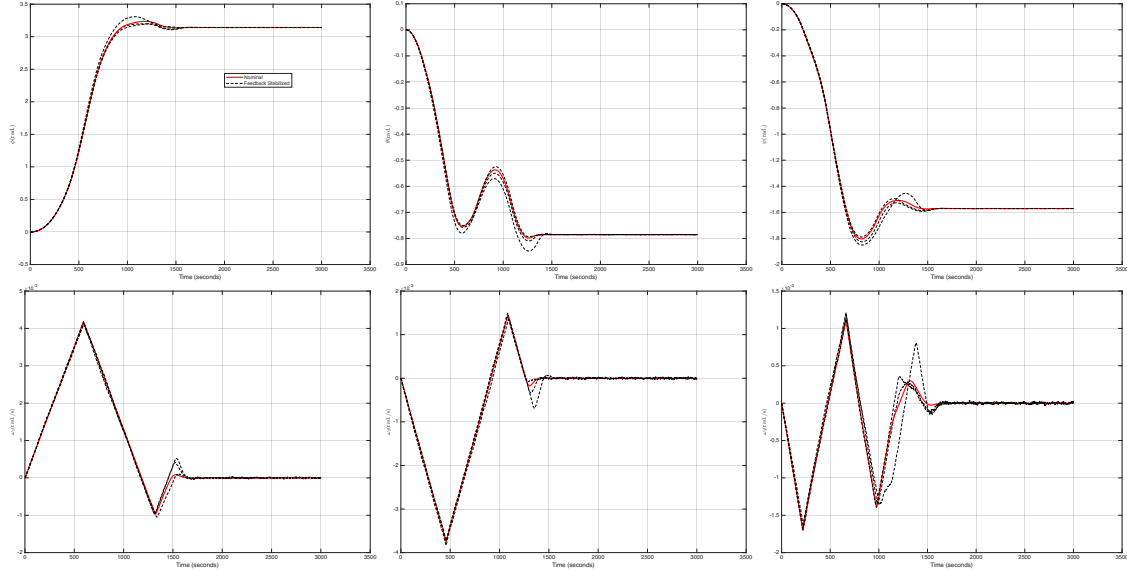


Figure 4: Closed-loop attitude response under disturbances and measurement noise. The nominal trajectory is shown in red, and three representative perturbed trajectories are shown using dotted black curves. The Box-iLQR feedback policy stabilizes the disturbed trajectories about the nominal maneuver.

Fuel-Optimal Orbit Transfer

Problem formulation The second example considers a planar low-thrust orbit-transfer problem with state

$$\mathbf{x} = [r \quad u \quad v]^T, \quad (42)$$

where r is the radial position, u is the radial velocity, and v is the transverse velocity. The controls are the normalized thrust magnitude s and the thrust-direction angle γ . The equations of motion are

$$\dot{r} = u, \quad (43)$$

$$\dot{u} = \frac{v^2}{r} - \frac{\mu}{r^2} + sa \sin \gamma, \quad (44)$$

$$\dot{v} = -\frac{uv}{r} + sa \cos \gamma, \quad (45)$$

where a is the maximum available acceleration. The controls satisfy

$$0 \leq s \leq \bar{s}, \quad -\frac{\pi}{3} \leq \gamma \leq \frac{\pi}{3}. \quad (46)$$

The initial conditions are

$$r(0) = 1.08, \quad u(0) = 0, \quad v(0) = \sqrt{\frac{\mu}{r(0)}}, \quad (47)$$

and the desired terminal conditions are

$$r(t_f) = 1.1, \quad u(t_f) = 0, \quad v(t_f) \text{ is free}, \quad t_f = 6\pi. \quad (48)$$

The original minimum-fuel objective is

$$J = \int_0^{t_f} s(t) dt. \quad (49)$$

Direct use of the nonsmooth absolute-value equivalent of this cost may produce poorly conditioned derivatives near zero thrust. Therefore, the running cost is replaced by the smooth approximation

$$\ell_s(s) = \sqrt{s^2 + \alpha^2} - \alpha, \quad \alpha = 10^{-12}. \quad (50)$$

For $|s| \gg \alpha$, this expression is effectively equal to $|s|$, while its first- and second-order derivatives remain well defined near $s = 0$.

The terminal equality constraints on r and u are incorporated through a quadratic terminal penalty. The discrete objective used by Box-iLQR is

$$J_h = \Delta t \sum_{k=0}^{N-1} \left(\sqrt{s_k^2 + \alpha^2} - \alpha + \frac{1}{2} \rho_\gamma \gamma^2 \right) + \frac{\rho_r}{2} (r_N - 1.1)^2 + \frac{\rho_u}{2} u_N^2, \quad (51)$$

where

$$\rho_r = 10^9, \quad \rho_u = 10^6, \quad \rho_\gamma = 5 \times 10^{-9}. \quad (52)$$

No terminal penalty is placed on v_N because the transverse velocity is free at the final time.

Nominal constrained solution The optimized thrust magnitude is shown in Fig. 5(a). The solution possesses a bang-off-bang-type structure containing three distinct active-thrust arcs separated by coast intervals. Each principal thrust arc lasts approximately 0.7 units of normalized time. This structure is consistent with the minimum-fuel nature of the objective, for which thrust is concentrated into relatively short intervals while remaining zero over much of the transfer.

The thrust-direction history is presented in Fig. 5(b). The control repeatedly approaches the direction bounds $\gamma = \pm\pi/3$, demonstrating that the direction constraint is active over portions of the maneuver. The evolution of the thrust profile as the control-barrier parameter is reduced is shown in Fig. 6. As the barrier parameter decreases, the solution approaches the limiting constrained thrust profile and the switching structure becomes increasingly distinct.

The radial position, radial velocity, and transverse velocity are shown in Fig. 7. The radial position completes approximately two and one-half oscillations with gradually increasing amplitude as it transitions from $r(0) = 1.08$ to the desired terminal value $r_f = 1.1$. The final radial position error is approximately

$$r(t_f) - r_f \approx -0.1115 \times 10^{-6}. \quad (53)$$

The radial velocity also displays an oscillatory response whose amplitude initially increases before decaying to approximately zero at the final time. The terminal radial-velocity error is

$$u(t_f) - u_f \approx -0.0429 \times 10^{-6}. \quad (54)$$

These small residuals demonstrate that the large quadratic terminal penalties provide an accurate numerical approximation of the original terminal equality constraints while remaining compatible with the unconstrained quadratic expansions used in iLQR.

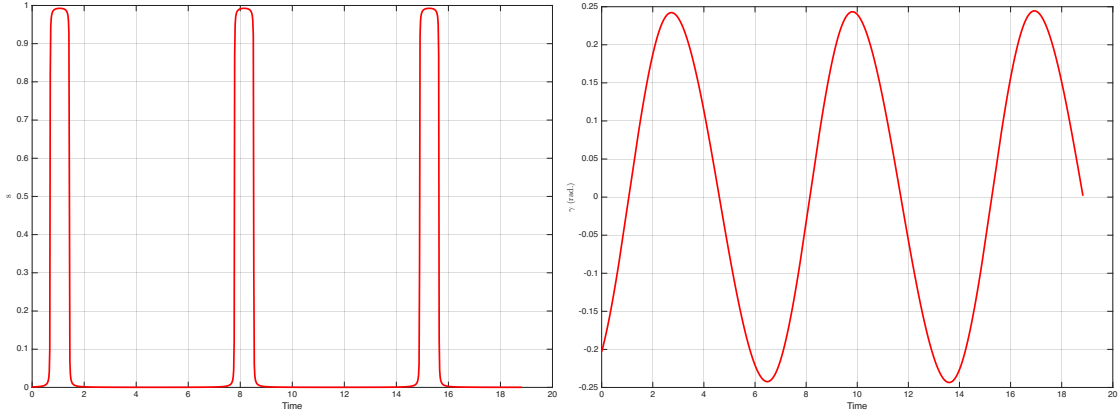


Figure 5: Optimal orbit-transfer controls: (a) normalized thrust magnitude, and (b) thrust-direction angle γ .

Closed-loop robustness to control uncertainty The robustness of the optimized transfer is evaluated by introducing uncertainty directly into the applied controls. The perturbed thrust-direction angle is modeled as

$$\tilde{\gamma}_k = \gamma_k + \delta\gamma_k, \quad \delta\gamma_k \sim \mathcal{N}(0, \sigma_\gamma^2), \quad \sigma_\gamma = 3^\circ = \frac{\pi}{60} \text{ rad.} \quad (55)$$

The applied thrust magnitude is modeled using a signal-dependent Gaussian perturbation,

$$\tilde{s}_k = s_k + \delta s_k, \quad \delta s_k \sim \mathcal{N}\left(0, (0.1s_k)^2\right). \quad (56)$$

The perturbed controls are projected onto their admissible bounds before being applied to the dynamics.

Figure 8 shows the radial-position and radial-velocity errors obtained when the noisy controls are applied without state feedback. Each point corresponds to one independent noise realization. The radial-position and radial-velocity errors are both of order 10^{-4} . Thus, relatively small control perturbations produce terminal errors that are several orders of magnitude larger than those obtained along the nominal trajectory.

Figure 9 shows the corresponding errors when the time-varying Box-iLQR feedback policy is applied. The radial-position error is reduced from order 10^{-4} to order 10^{-7} . Under the adopted normalization, this corresponds to a dimensional position error on the order of tens of meters. The radial-velocity error is reduced from order 10^{-4} to approximately order 10^{-6} . The results therefore show that the feedback matrices obtained as a byproduct of the Box-iLQR backward pass provide substantial robustness without requiring a separate controller-design step.

Figure 10 compares the actual fuel-consumption metric, $\int_0^{t_f} s(t) dt$, for the nominal optimal trajectory and the perturbed trajectories after feedback correction. This quantity is distinct from the smoothed objective used internally by Box-iLQR. The solid black line represents the nominal optimal fuel cost of 2.2228, while each red marker represents the final fuel cost obtained for an independent perturbation realization. For all realizations, the corrected fuel cost remains less than 1% above the nominal optimal value, indicating that the feedback correction compensates for the applied uncertainty while introducing only a small increase in actual fuel consumption.

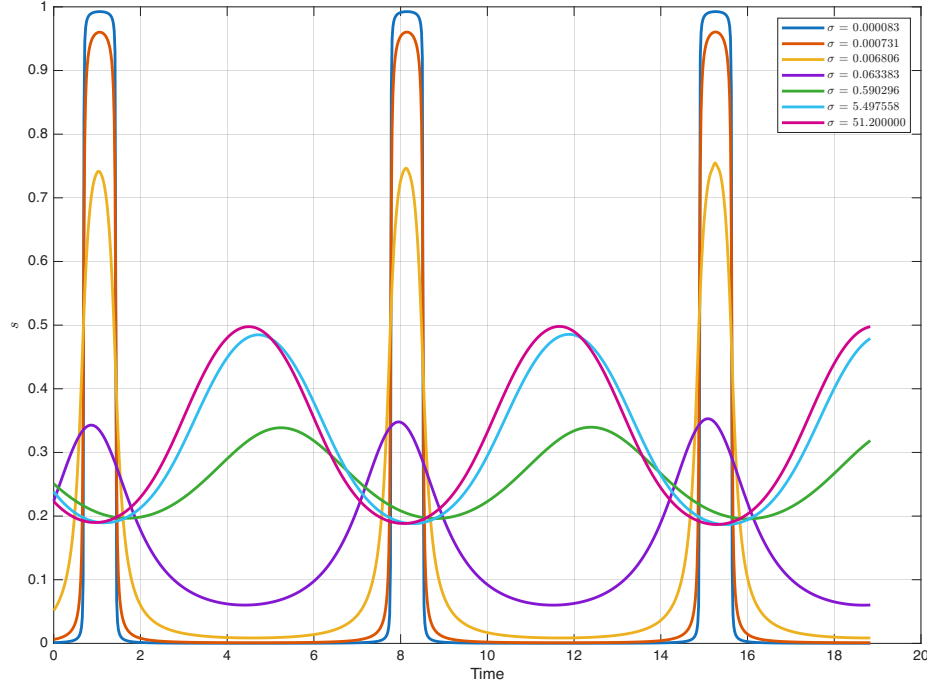


Figure 6: Evolution of the thrust history for various values of the log-barrier parameter.

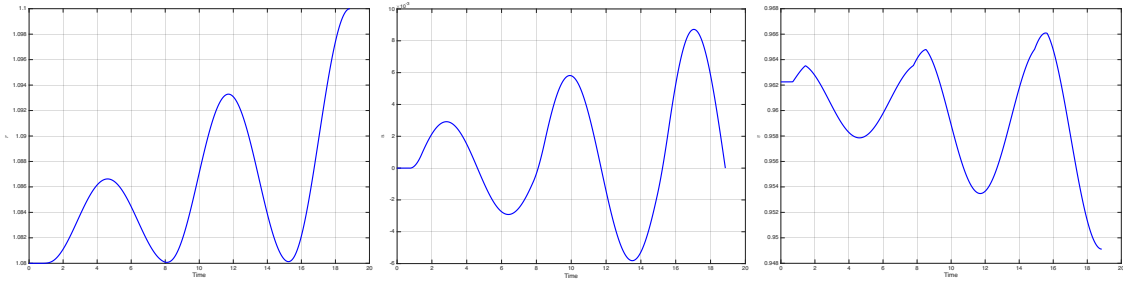


Figure 7: Nominal orbit-transfer state histories: radial position r , radial velocity u , and transverse velocity v .

The two numerical examples demonstrate complementary properties of the proposed approach. In the attitude-control problem, the optimal solution contains several long-duration control-boundary arcs, yet Box-iLQR generates a smoothly convergent state trajectory and a feedback policy that stabilizes the maneuver under simultaneous disturbance torques and measurement noise. In the orbit-transfer problem, the method recovers the expected sparse bang–off–bang thrust structure while respecting the thrust-direction bounds. The terminal quadratic penalty enforces the desired final radius and radial velocity to high accuracy, and the smoothed fuel objective avoids the derivative singularity associated with the nonsmooth minimum-fuel cost.

Most importantly, the time-varying feedback gains are obtained directly during the Box-iLQR backward pass. No independent linearization or secondary controller-synthesis procedure is required. The Monte Carlo and control-noise experiments show that these gains substantially reduce trajectory and terminal errors relative to open-loop execution. Thus, the proposed framework pro-

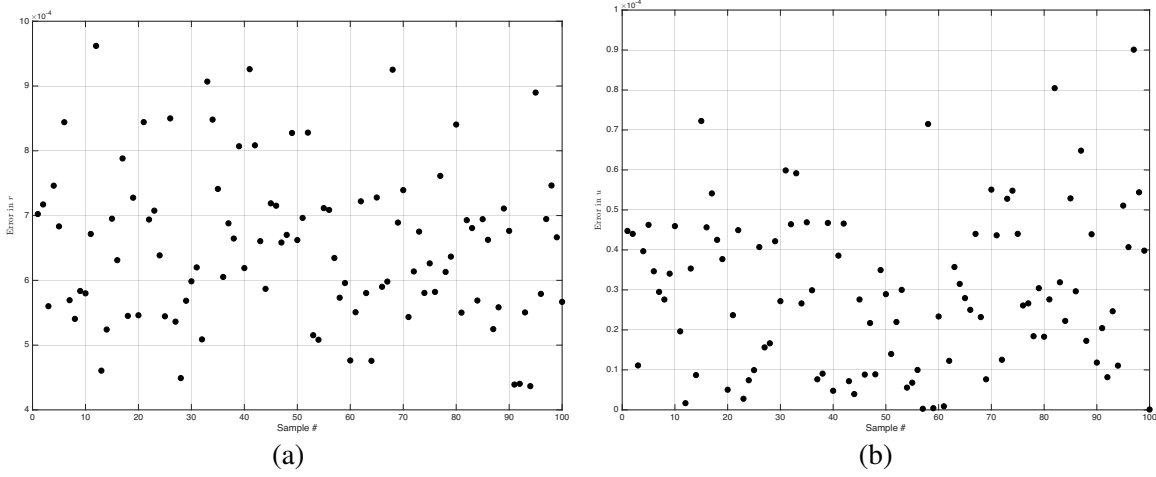


Figure 8: Samplewise terminal errors for the orbit-transfer problem without feedback: (a) radial-position error and (b) radial-velocity error. The horizontal axis denotes the noise-realization index.

vides both a constrained nominal solution and a computationally inexpensive closed-loop policy for robust trajectory execution.

Fixed-Time L_2 -to- L_2 Halo-Orbit Transfer

The next numerical study considers a transfer between two southern L_2 Halo orbits in the Earth–Moon system. The boundary conditions, transfer time, and reference transfer are adopted from Refs.^{19–21} In contrast to the impulsive and trajectory-correction formulations in these references, the maneuver considered here is solved directly as a continuous, control-constrained orbit-transfer problem. The final time is fixed at

$$t_f = 13.0789122308 \text{ TU},$$

and is not included as an optimization variable because free-final-time optimization is beyond the scope of the present work.

The spacecraft dynamics are modeled using the circular restricted three-body problem (CR3BP). The state and control vectors are

$$\mathbf{x} = [x \ y \ z \ \dot{x} \ \dot{y} \ \dot{z}]^T, \quad \mathbf{u} = [u_x \ u_y \ u_z]^T,$$

where position is expressed in length units (LU), velocity in velocity units (VU), time in time units (TU), and control acceleration in LU/TU^2 . The controlled CR3BP equations are

$$\ddot{x} - 2\dot{y} = \frac{\partial \Omega}{\partial x} + u_x, \quad (57)$$

$$\ddot{y} + 2\dot{x} = \frac{\partial \Omega}{\partial y} + u_y, \quad (58)$$

$$\ddot{z} = \frac{\partial \Omega}{\partial z} + u_z, \quad (59)$$

where

$$\Omega(x, y, z) = \frac{1 - \mu}{d_1} + \frac{\mu}{d_2} + \frac{1}{2} (x^2 + y^2),$$

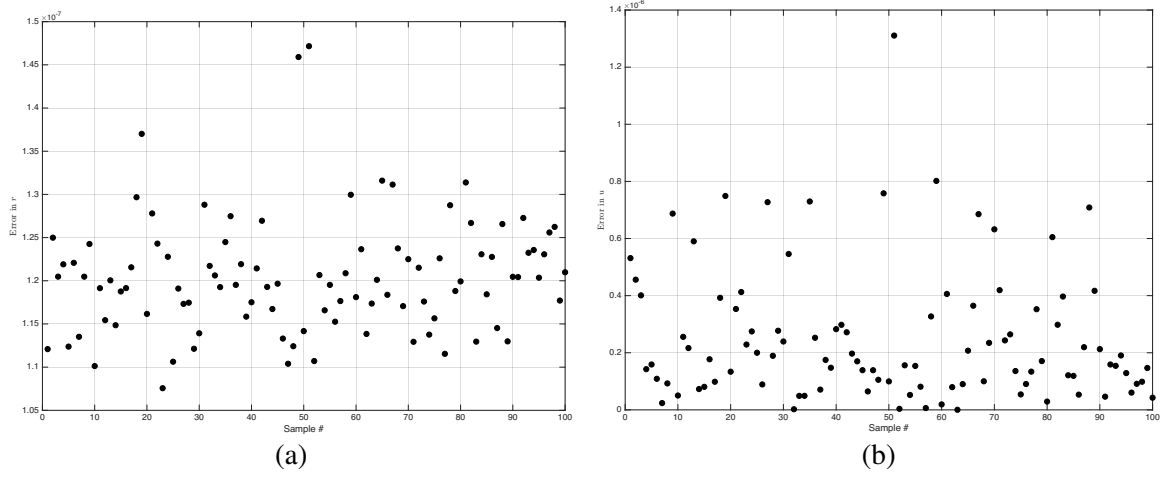


Figure 9: Samplewise terminal errors for the orbit-transfer problem with Box-iLQR feedback: (a) radial-position error and (b) radial-velocity error. The feedback policy reduces the respective errors to approximately orders 10^{-7} and 10^{-6} .

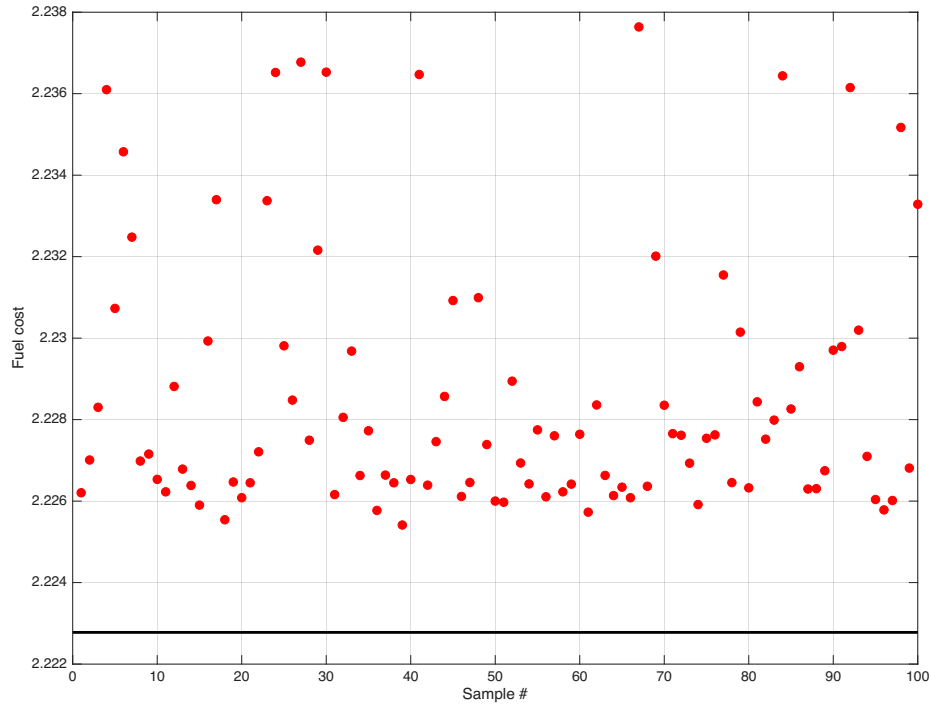


Figure 10: Comparison of the nominal optimal fuel cost and the fuel costs obtained after perturbation and feedback correction. The quantity shown is the actual fuel-consumption metric, $\int_0^{t_f} s(t) dt$, rather than the smoothed objective used internally by the optimization algorithm. The solid black line denotes the nominal optimal fuel cost, $\int_0^{t_f} s^*(t) dt = 2.2228$, while the red markers denote the corrected fuel cost for each independent perturbation realization. The horizontal axis indicates the sample number.

with

$$d_1 = \sqrt{(x + \mu)^2 + y^2 + z^2}, \quad d_2 = \sqrt{(x - 1 + \mu)^2 + y^2 + z^2}.$$

Each control component is constrained according to

$$-\frac{0.05}{\sqrt{3}} \leq u_i(t) \leq \frac{0.05}{\sqrt{3}}, \quad i \in \{x, y, z\}. \quad (60)$$

To approximate the minimum- Δv objective while retaining smooth first- and second-order derivatives, the running cost is selected as

$$\ell(\mathbf{u}) = 100 \left(\sqrt{u_x^2 + u_y^2 + u_z^2 + \alpha^2} - \alpha \right), \quad \alpha = 10^{-6}. \quad (61)$$

For $\|\mathbf{u}\| \gg \alpha$, this expression approaches $100\|\mathbf{u}\|$, while remaining twice differentiable near $\mathbf{u} = \mathbf{0}$. The complete objective is

$$J = \frac{1}{2} (\mathbf{x}(t_f) - \mathbf{x}_f)^T Q_f (\mathbf{x}(t_f) - \mathbf{x}_f) + \int_0^{t_f} \ell(\mathbf{u}(t)) dt, \quad (62)$$

where

$$Q_f = 50 \times 10^4 \text{diag} (10^4, 10^4, 10^4, 1, 1, 1). \quad (63)$$

The terminal condition is therefore imposed through a quadratic penalty rather than as an exact terminal equality constraint. Consequently, a small nonzero terminal residual remains even along the nominal optimized trajectory.

Figure 11 shows the reduction of the control-barrier parameter over the Box-iLQR iterations. A total of 172 outer iterations are performed. Two solutions from this continuation sequence are of particular interest. At iteration 120, the barrier parameter is $\sigma = 0.0401$, whereas the final iteration corresponds to $\sigma = 4.945 \times 10^{-6}$. The latter gives a closer approximation to the fixed-time fuel-optimal constrained solution, while the former retains more interior control authority and produces substantially more effective feedback gains.

The control histories obtained at the final barrier value are shown in Fig. 12. All three components contain extended zero-control intervals, which are consistent with the minimum-fuel objective. The u_y and u_z components also contain arcs along their respective control bounds, whereas u_x remains strictly within its admissible interval. After removing the factor of 100 in Eq. (61) and converting the resulting normalized velocity from LU/TU to dimensional units, the maneuver cost is

$$\Delta v_{\text{final}} = 283.7073 \text{ m/s}.$$

The corresponding controls at iteration 120 are shown in Fig. 13. These controls also contain intervals over which the thrust is zero; however, none of the three components reaches its saturation boundary. The dimensional maneuver cost is

$$\Delta v_{120} = 287.1338 \text{ m/s},$$

which is only approximately 1.21% larger than the final low-barrier cost. Thus, the iteration-120 trajectory sacrifices only a small amount of nominal fuel optimality while preserving appreciable control margin for closed-loop correction.

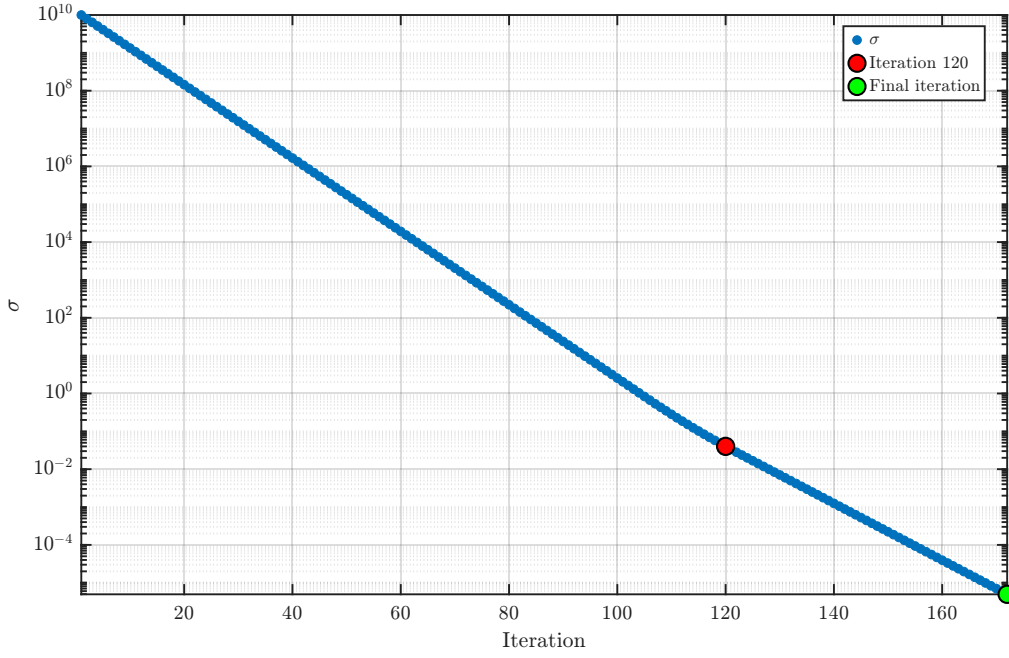


Figure 11: Reduction of the control log-barrier parameter during the fixed-time CR3BP transfer. Iteration 120 corresponds to $\sigma = 0.0401$, and iteration 172 corresponds to $\sigma = 4.945 \times 10^{-6}$.

The state histories corresponding to the two barrier values are compared in Fig. 14. Although the two solutions have substantially different closed-loop robustness properties, their nominal position and velocity trajectories are nearly indistinguishable. In all six state components, the trajectory obtained at $\sigma = 0.0401$ closely follows the trajectory obtained at $\sigma = 4.945 \times 10^{-6}$, with the two curves overlapping over nearly the entire transfer. Thus, stopping the barrier continuation at iteration 120 causes only a negligible change in the nominal state evolution while preserving significantly greater control authority for feedback correction.

Because a terminal quadratic penalty is used, neither nominal solution reaches the prescribed final state exactly. For $\sigma = 0.0401$, the dimensional terminal residual is

$$\mathbf{e}_{f,120} = \begin{bmatrix} 13.3345 \\ 4.5538 \\ 2.6976 \\ -0.1365 \\ 0.2022 \\ -0.0222 \end{bmatrix}, \quad (64)$$

where the first three components are expressed in meters and the final three components are expressed in meters per second. The corresponding position- and velocity-error norms are

$$\|\mathbf{e}_{r,f,120}\| = 14.3465 \text{ m}, \quad \|\mathbf{e}_{v,f,120}\| = 0.2450 \text{ m/s}.$$

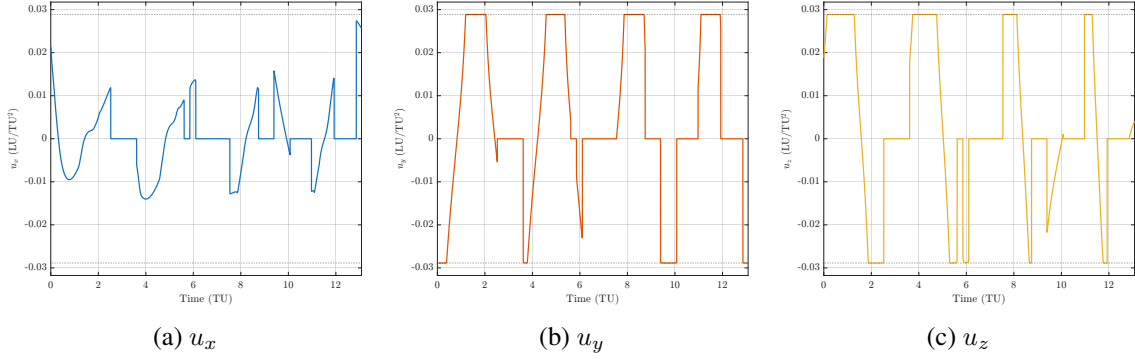


Figure 12: Control histories at the final barrier value $\sigma = 4.945 \times 10^{-6}$. The dotted black lines denote the componentwise bounds $\pm 0.05/\sqrt{3}$ LU/TU². The u_y and u_z controls contain saturated arcs, while all three components contain coast intervals.

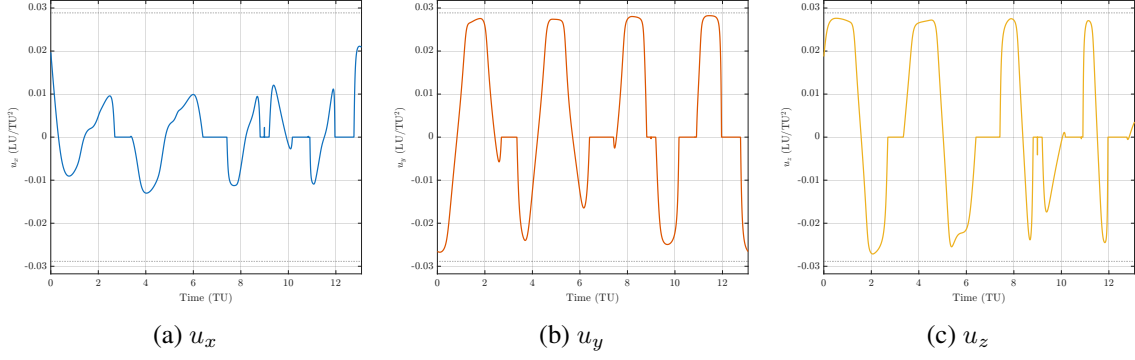


Figure 13: Control histories at iteration 120, corresponding to $\sigma = 0.0401$. The solution contains zero-control intervals but remains strictly interior to the componentwise control bounds.

For $\sigma = 4.945 \times 10^{-6}$, the terminal residual is

$$\mathbf{e}_{f,\text{final}} = \begin{bmatrix} 12.7258 \\ 4.4417 \\ 2.4579 \\ -0.1353 \\ 0.1866 \\ -0.0213 \end{bmatrix}, \quad (65)$$

with

$$\|\mathbf{e}_{r,f,\text{final}}\| = 13.7009 \text{ m}, \quad \|\mathbf{e}_{v,f,\text{final}}\| = 0.2315 \text{ m/s}.$$

The smaller residuals at the final barrier value are consistent with its slightly lower nominal objective, although the difference between the two solutions is small compared with their substantially different closed-loop behavior.

To evaluate robustness, 100 independent Monte Carlo simulations are performed. Instead of applying the nominal control directly, each component is perturbed according to the signal-dependent

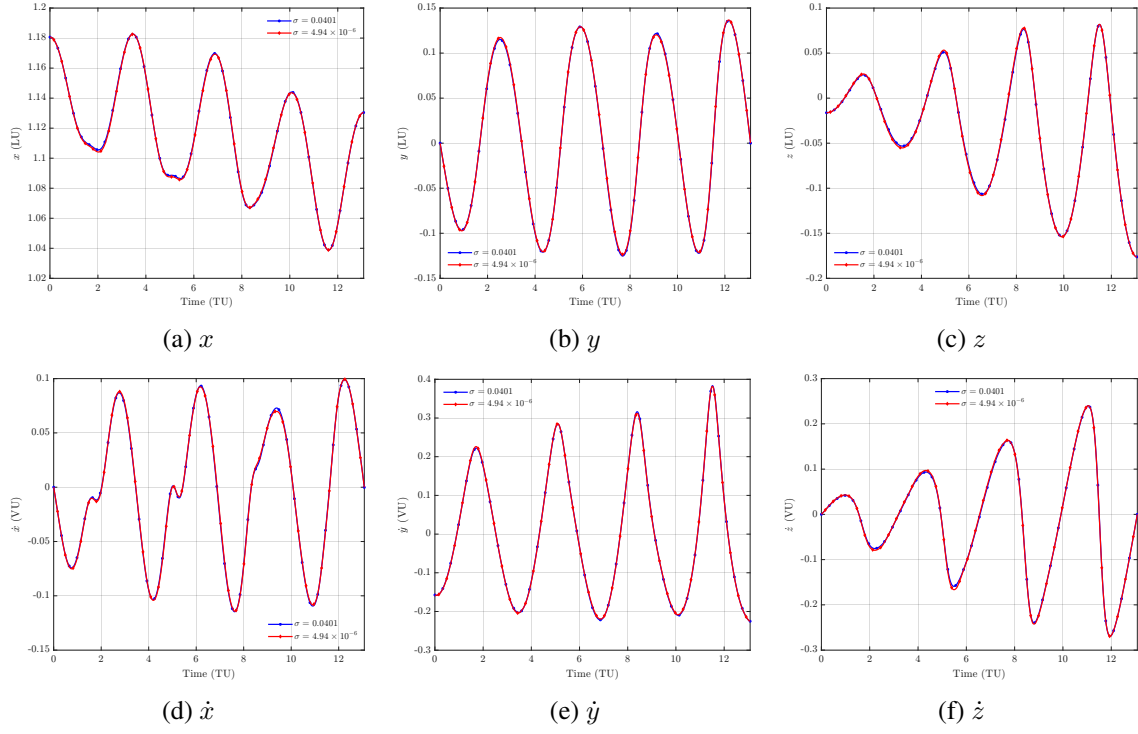


Figure 14: Comparison of the nominal CR3BP state histories obtained at $\sigma = 0.0401$ and $\sigma = 4.945 \times 10^{-6}$. The solution at $\sigma = 0.0401$ is shown using blue lines with circular markers, whereas the solution at $\sigma = 4.945 \times 10^{-6}$ is shown using red lines with diamond markers. The position components are expressed in LU and the velocity components are expressed in VU. The curves nearly overlap, indicating that the two barrier values produce almost identical nominal state trajectories despite their substantially different closed-loop robustness.

Gaussian model

$$\tilde{u}_{i,k} \sim \mathcal{N} \left(\bar{u}_{i,k}, \left(\frac{|\bar{u}_{i,k}|}{30} \right)^2 \right), \quad i \in \{x, y, z\}. \quad (66)$$

Thus, the 3σ uncertainty in each component is equal to 10% of the magnitude of that nominal control component. The noisy control is corrected using the time-varying feedback gains obtained from the same Box-iLQR iteration,

$$\mathbf{u}_k^{\text{cl}} = \Pi_{\mathcal{U}} \left[\tilde{\mathbf{u}}_k + K_k \left(\mathbf{x}_k^{\text{noisy}} - \bar{\mathbf{x}}_k \right) \right], \quad (67)$$

where $\Pi_{\mathcal{U}}[\cdot]$ denotes projection onto the control box in Eq. (60). Figure 15 shows the resulting controls for the iteration-120 feedback policy. The solid black curves denote the nominal controls, and the gray curves denote the 100 noisy closed-loop realizations.

Table 2 summarizes the dimensional terminal-error statistics. Without feedback, the control perturbations lead to complete departure from the nominal transfer, with position errors on the order of 10^9 m. The feedback gains obtained at $\sigma = 4.945 \times 10^{-6}$ also perform poorly, producing position errors on the order of 10^8 m and velocity errors on the order of 10^3 m/s. In contrast, the feedback gains obtained at $\sigma = 0.0401$ reduce the mean and maximum position-error norms to 20.1178

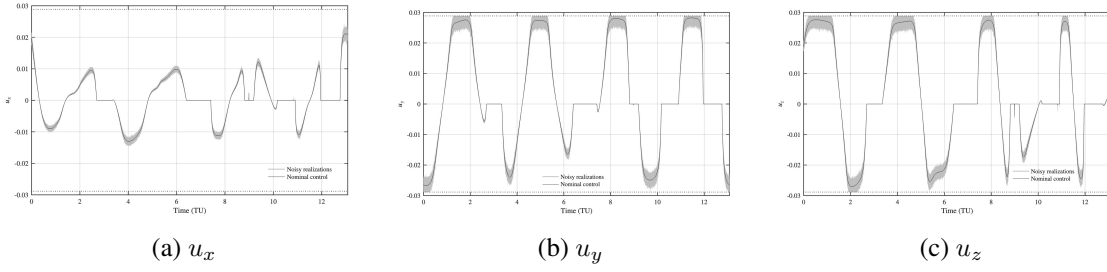


Figure 15: Closed-loop control histories for 100 Monte Carlo realizations using the feedback gains obtained at $\sigma = 0.0401$. The nominal controls are shown using solid black curves, and the noisy feedback-corrected controls are shown in gray.

Table 2: Terminal-error norms for 100 noisy-control realizations.

Case	Mean position error (m)	Maximum position error (m)	Mean velocity error (m/s)	Maximum velocity error (m/s)
No feedback, $\sigma = 0.0401$	1.0840×10^9	1.7167×10^9	8.9771×10^3	6.6314×10^5
Feedback, $\sigma = 0.0401$	20.1178	55.4886	0.2457	0.2523
Feedback, $\sigma = 4.945 \times 10^{-6}$	4.9669×10^8	5.5826×10^8	1.5214×10^3	1.7420×10^3

m and 55.4886 m, respectively. The corresponding mean and maximum velocity-error norms are 0.2457 m/s and 0.2523 m/s.

The maximum maneuver cost among the 100 feedback-corrected realizations at $\sigma = 0.0401$ is

$$\Delta v_{\max} = 287.3046 \text{ m/s.}$$

This is only 0.0595% larger than the nominal iteration-120 cost and approximately 1.27% larger than the final low-barrier cost. Hence, the feedback correction produces a negligible increase in fuel consumption while substantially improving terminal accuracy.

For comparison, Yamamoto et al.²¹ apply NMPC corrections over the transfer segments and include an additional orbital period after the final impulse to capture the target Halo orbit. Their Monte Carlo analysis reports a maximum terminal miss distance of approximately 0.6101 km. The median and maximum correction costs are 308.54 m/s and 409.93 m/s, respectively, compared with a reference impulsive cost of 202.2024 m/s. Thus, their maximum corrected cost is approximately twice the reference cost. In the present study, the maximum position error is only 55.4886 m, despite the absence of an additional target-orbit correction period, and the increase in Δv relative to the selected nominal trajectory is negligible. Although the uncertainty models and correction architectures are not identical, the comparison demonstrates the strong closed-loop performance obtained from the Box-iLQR feedback policy.

These results also reveal an important distinction between nominal optimality and closed-loop robustness. Reducing the barrier parameter to $\sigma = 4.945 \times 10^{-6}$ drives the nominal control closer to the fuel-optimal constrained solution, but it also produces saturated arcs in u_y and u_z . Near an active control boundary, the curvature of the logarithmic barrier increases and the corresponding feedback channel is attenuated. This constraint-aware behavior prevents the feedback law from

aggressively commanding an already saturated actuator, but it also leaves little remaining control authority for disturbance rejection. In contrast, the solution at $\sigma = 0.0401$ remains interior to the control box. Its feedback gains therefore retain sufficient authority to correct the noisy trajectory while still respecting the prescribed control limits.

This behavior is consistent with the constraint-aware feedback analysis in Ref.¹⁸, where the logarithmic barrier curvature is shown to encode proximity to state and control constraints directly into the iLQR feedback matrices. For the present CR3BP transfer, stopping the barrier continuation before reaching its smallest value produces a slightly suboptimal nominal maneuver but a dramatically superior closed-loop policy. The result demonstrates that logarithmic barriers provide more than a mechanism for enforcing control constraints: they also expose a useful tradeoff between open-loop fuel optimality, available control margin, and closed-loop robustness.

CONCLUSIONS AND FUTURE WORK

This paper presented Box-iLQR, a logarithmic-barrier-based extension of the iterative Linear Quadratic Regulator for nonlinear optimal-control problems subject to box constraints on the states and control inputs. The state and control constraints were incorporated directly into the running and terminal costs using logarithmic barrier functions, thereby converting the constrained problem into a sequence of smooth unconstrained subproblems. In addition to maintaining strict feasibility, the barrier Hessians contributed positive curvature to the local quadratic approximations and improved the numerical conditioning of the iLQR backward pass. An adaptive outer-loop procedure was used to reduce the barrier parameters, while the solution of each subproblem provided a warm start for the subsequent barrier problem.

The method was demonstrated on two representative spacecraft optimal-control problems. For the fully actuated spacecraft attitude maneuver, Box-iLQR generated a feasible trajectory containing extended control-saturation arcs while driving the Euler angles and angular velocities to their desired terminal values. The time-varying feedback gains obtained from the converged backward pass were subsequently used for closed-loop trajectory tracking. Numerical simulations showed that these gains stabilized the maneuver in the presence of disturbance torques, sinusoidal disturbances, and measurement noise. For the minimum-fuel orbit-transfer problem, a smooth approximation of the fuel cost was employed together with bounds on the thrust magnitude and thrust-direction angle. The resulting solution exhibited the expected bang-off-bang structure and satisfied the desired terminal radius and radial velocity to high accuracy. The closed-loop simulations further showed that the Box-iLQR feedback policy substantially reduced the terminal errors caused by thrust-magnitude and thrust-direction uncertainty. These results demonstrate that Box-iLQR provides both a constrained nominal trajectory and a locally stabilizing feedback policy within a single computational framework.

Future work will focus on three extensions. First, we will develop an inexact continuation strategy that reduces the logarithmic-barrier parameters before each intermediate subproblem has fully converged, thereby lowering the overall computational cost. Second, terminal boundary conditions will be imposed directly as equality constraints instead of being approximated using large quadratic terminal penalties. Finally, Box-iLQR will be extended to free-final-time problems so that the maneuver duration can be optimized together with the state and control trajectories, as required in many orbit-transfer applications.

REFERENCES

- [1] J. T. Betts, "Survey of Numerical Methods for Trajectory Optimization," *Journal of Guidance, Control, and Dynamics*, Vol. 21, No. 2, 1998, pp. 193–207, 10.2514/2.4231.
- [2] A. V. Rao, "A Survey of Numerical Methods for Optimal Control," *Advances in the Astronautical Sciences*, Vol. 135, No. 1, 2009, pp. 497–528. AAS Paper 09-334.
- [3] M. Kelly, "An Introduction to Trajectory Optimization: How to Do Your Own Direct Collocation," *SIAM Review*, Vol. 59, No. 4, 2017, pp. 849–904, 10.1137/16M1062569.
- [4] D. Malyuta, Y. Yu, P. Elango, and B. Açıkmeşe, "Advances in Trajectory Optimization for Space Vehicle Control," *Annual Reviews in Control*, Vol. 52, 2021, pp. 282–315, 10.1016/j.arcontrol.2021.04.013.
- [5] S. R. Vadali and J. L. Junkins, "Optimal Open-Loop and Stable Feedback Control of Rigid Spacecraft Attitude Maneuvers," *The Journal of the Astronautical Sciences*, Vol. 32, No. 2, 1984, pp. 105–122.
- [6] C. K. Carrington and J. L. Junkins, "Optimal Nonlinear Feedback Control for Spacecraft Attitude Maneuvers," *Journal of Guidance, Control, and Dynamics*, Vol. 9, No. 1, 1986, pp. 99–107, 10.2514/3.20073.
- [7] R. M. Byers and S. R. Vadali, "Quasi-Closed-Form Solution to the Time-Optimal Rigid Spacecraft Reorientation Problem," *Journal of Guidance, Control, and Dynamics*, Vol. 16, No. 3, 1993, pp. 453–461, 10.2514/3.21031.
- [8] J. L. Crassidis, S. R. Vadali, and F. L. Markley, "Optimal Variable-Structure Control Tracking of Spacecraft Maneuvers," *Journal of Guidance, Control, and Dynamics*, Vol. 23, No. 3, 2000, pp. 564–566, 10.2514/2.4568.
- [9] K. S. Phogat, D. Chatterjee, and R. N. Banavar, "Discrete-Time Optimal Attitude Control of a Spacecraft with Momentum and Control Constraints," *Journal of Guidance, Control, and Dynamics*, Vol. 41, No. 1, 2018, pp. 199–211, 10.2514/1.G002861.
- [10] T. Haberkorn, P. Martinon, and J. Gergaud, "Low-Thrust Minimum-Fuel Orbital Transfer: A Homotopic Approach," *Journal of Guidance, Control, and Dynamics*, Vol. 27, No. 6, 2004, pp. 1046–1060, 10.2514/1.4022.
- [11] I. M. Ross, Q. Gong, and P. Sekhavat, "Low-Thrust, High-Accuracy Trajectory Optimization," *Journal of Guidance, Control, and Dynamics*, Vol. 30, No. 4, 2007, pp. 921–933, 10.2514/1.23181.
- [12] E. Taheri, I. Kolmanovsky, and E. M. Atkins, "Enhanced Smoothing Technique for Indirect Optimization of Minimum-Fuel Low-Thrust Trajectories," *Journal of Guidance, Control, and Dynamics*, Vol. 39, No. 11, 2016, pp. 2500–2511, 10.2514/1.G000379.
- [13] D. Q. Mayne, "A Second-Order Gradient Method for Determining Optimal Trajectories of Non-Linear Discrete-Time Systems," *International Journal of Control*, Vol. 3, No. 1, 1966, pp. 85–95, 10.1080/00207176608921369.
- [14] W. Li and E. Todorov, "Iterative Linear Quadratic Regulator Design for Nonlinear Biological Movement Systems," *Proceedings of the First International Conference on Informatics in Control, Automation and Robotics*, Vol. 1, SciTePress, 2004, pp. 222–229, 10.5220/0001143902220229.
- [15] Y. Tassa, T. Erez, and E. Todorov, "Synthesis and Stabilization of Complex Behaviors through Online Trajectory Optimization," *2012 IEEE/RSJ International Conference on Intelligent Robots and Systems*, IEEE, 2012, pp. 4906–4913, 10.1109/IROS.2012.6386025.
- [16] Y. Tassa, N. Mansard, and E. Todorov, "Control-Limited Differential Dynamic Programming," *2014 IEEE International Conference on Robotics and Automation*, IEEE, 2014, pp. 1168–1175, 10.1109/ICRA.2014.6907001.
- [17] J. Nocedal and S. J. Wright, *Numerical Optimization*. New York: Springer, 2 ed., 2006, 10.1007/978-0-387-40065-5.
- [18] Abhijeet and S. Chakravorty, "Safe Optimal Control using Log Barrier Constrained iLQR," 2026.
- [19] K. Saloglu and E. Taheri, "Acceleration-Based Switching Surfaces for Impulsive Trajectory Design Between Cislunar Libration Point Orbits," *The Journal of the Astronautical Sciences*, Vol. 71, No. 2, 2024, p. 13, 10.1007/s40295-024-00432-z.
- [20] K. Yamamoto, E. Taheri, and J. L. Junkins, "Jacobian Coloring for Multi-Impulse Cislunar Trajectory Optimization," SSRN preprint, Sept. 2025. SSRN 5545742, 10.2139/ssrn.5545742.
- [21] K. Yamamoto, E. Taheri, and J. L. Junkins, "Nonlinear MPC for Cislunar Trajectory Correction under Impulsive Maneuver Uncertainties: A Computational Performance Analysis," *Proceedings of the 2026 American Control Conference (ACC)*, New Orleans, Louisiana, USA, May 2026. Paper WeC04.4.